Checkpointing

Compiler Sessional Online

Problem Statement

Create two commands -

Checkpoint (`C`) This creates a checkpoint of the current context.

Restore (`R`) This is used to restore the context to the last checkpoint.

When a checkpoint is restored, any changes made after the last checkpoint, is reverted. No memory leaks or crashes will be tolerated.

You can consider the checkpoints (`C`) and restores (`R`) as pairs. A restoration will cause the context to revert back to the paired checkpoint.

There can be unmatched checkpoints (a checkpoint created, but never restored to), but there can't be unmatched restores (an attempt to restore without ever creating a checkpoint). Such unmatched restores will cause an error.

Refer to the sample input and output and their explanation for clarification.

Restrictions

- You **can** modify existing classes from your practice assignment
- You **can** use STL
- Must compile with `-fsanitize=address`

Examples

Sample Input

```
3
R
I a int
C
S
I b float
C
I c double
R
L c
L b
R
S
P A
R
Q
```

Sample Output

```
 ScopeTable# 1 created
Cmd 1: R
 ERROR: No checkpoints to restore to
Cmd 2: I a int
 Inserted in ScopeTable# 1 at position 2, 1
Cmd 3: C
 Checkpoint created
Cmd 4: S
 ScopeTable# 2 created
Cmd 5: I b float
 Inserted in ScopeTable# 2 at position 3, 1
Cmd 6: C
 Checkpoint created
Cmd 7: I c double
 Inserted in ScopeTable# 2 at position 1, 1
Cmd 8: R
 Restored to last checkpoint
Cmd 9: L c
 'c' not found in any of the ScopeTables
Cmd 10: L b
 'b' found in ScopeTable# 2 at position 3, 1
Cmd 11: R
 Restored to last checkpoint
Cmd 12: S
 ScopeTable# 2 created
Cmd 13: P A
 ScopeTable# 2
 1-->
 2-->
 3-->
  ScopeTable# 1
  1-->
  2--> <a,int>
  3-->
Cmd 14: R
 ERROR: No checkpoints to restore to
Cmd 15: Q
 ScopeTable# 2 removed
 ScopeTable# 1 removed
```

Explanation

1. Command 1 tried to restore, but there was no checkpoints to restore to. So it raised an error.

2. Command 3 and Command 6 created checkpoints

3. Notice that `C` in command 3 is paired with `R` in command 11. Similarly `C` in command 6 is paired with `R` in command 8.

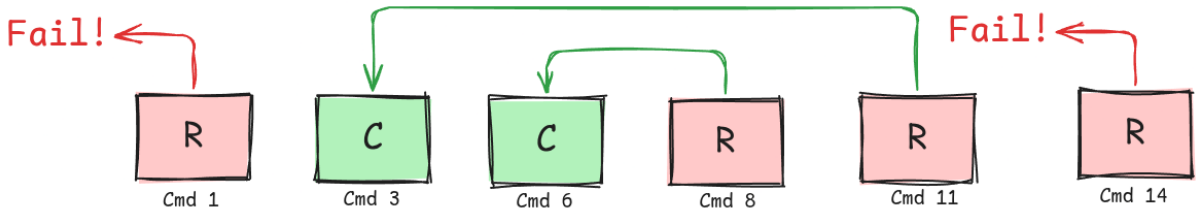


Figure 1: Sample input scenario

4. Command 8 reverted the context back to `C` in command 6. This restoration deleted the symbol `c` inserted in command 7.
5. Because `c` was deleted, the lookup failed in command 9
6. Command 11 reverted the context back to `C` in command 3. This restoration deleted the symbol `b` inserted in command 5 along with the scope created in command 4
7. Command 14 does not have a matching checkpoint
8. Also note that the scope created in command 12 reused the ID 2, because of the restoration in command 11.

Marks Distribution

Task	**Marks**
Checkpoint saving	5
Restoration	5
Total	**10**